

LAB # 4**DISCRETE TIME CONVOLUTION****OBJECTIVE:**

The basic objective of this lab is to perform the convolution in Matlab.

1. DISCRETE TIME CONVOLUTION:

An LTI discrete-time system is (equivalently to the continuous-time case) completely describes by impulse response, which is usually denoted by $h[n]$. The impulse response of an LTI discrete-time system is the output of the system when the unit impulse sequence is applied as input. Even though the unit impulse input consists of one non-zero term, the impulse response signal $h[n]$ usually consists of more than one non-zero elements. The explanation is that the system is dynamic (with memory); that is, the system responds over various time instances to the output applied at $n = 0$. The knowledge of the impulse response $h[n]$ of a system allows the computation of the response $y[n]$ of the system to any input signal $x[n]$. The output signal is computed by discrete time convolution. The mathematical expression of discrete time convolution is

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k] = \sum_{k=-\infty}^{+\infty} h[k]x[n-k]$$

The convolution between two discrete time signals is computed by using the MATLAB command conv.

2.1 COMPUTATION OF DISCRETE TIME CONVOLUTION:

The process that has to be followed to analytically derive the convolution sum is as follows. First, the input and impulse response signals are plotted in the k -axis. One of the two signals is reversed about the amplitude axis and its reflection is shifted from $-\infty$ to $+\infty$ by changing appropriately the value of n . The output of the system is computed from the overlapping values of $x[n]$ and $h[n-k]$ according to the convolution sum

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k].$$

There are three basic principles that should be considered while computing the convolution between two discrete time signals $x[n]$ and $h[n]$ using conv command as follows:

$$y = \text{conv}(x, h)$$

1. The length and time interval of input and impulse response should be same.
2. There should be no overlap between interval for input and impulse response signal.
3. The length of output vector 'y' should be calculated by following formulae:

$$\text{length}(y) = \text{length}(x) + \text{length}(h) - 1.$$

2. If non zeros values of $x[n]$ are in the interval $[a_x, b_x]$ and non zero values of $h[n]$ are in the Interval $[a_h, b_h]$ then the non zero values of the output $y[n]$ are in the interval $[a_x + a_h, b_x + b_h]$.

2.2 CONVOLUTION AS SUM OF SHIFTED AND SCALED IMPULSE RESPONSE:

- For any given n, one point convolution is evaluated as
 - Step 1: time reversal of either signal (e.g., $x[k] \rightarrow x[-k]$ or $h[k] \rightarrow h[-k]$)
 - Step 2: shift $h[-k]$ by n samples to obtain $h[n-k]$
 - Step 3: multiply $x[k]$ and $h[n-k]$ for each k and then take the summation over k

2.3 CONVOLUTION USING LOOPS

Convolution can also be evaluated by using loops and if else statement.

Lab Tasks

In-Lab Task 01: Compute convolution of following signals by using 'conv' command and loops

$$x[n] = \delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases} \quad h[n] = [2 \ 4 \ 1 \ 3], 0 \leq n \leq 3$$

Post-Lab Task 01: Compute convolution of following signals by using 'conv' command and loops and as sum of shifted and scaled impulse response(take help from lab 2)

$$x[n] = \begin{cases} 1 & 0 \leq n \leq 4 \\ 0 & \text{elsewhere} \end{cases}$$

and

$$h[n] = \begin{cases} 1.5^n & 0 \leq n \leq 6 \\ 0 & \text{elsewhere} \end{cases}$$